

GPS-BASED EDUTRACK PACK



UR6523002

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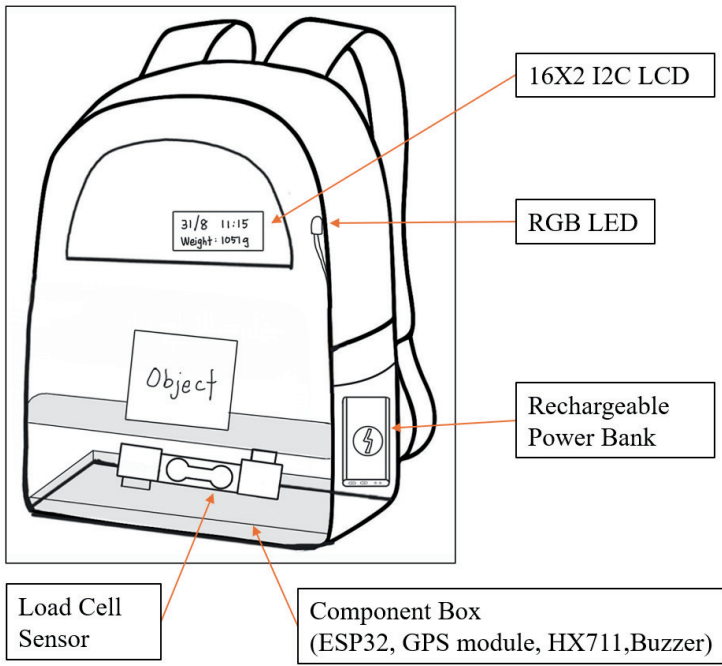
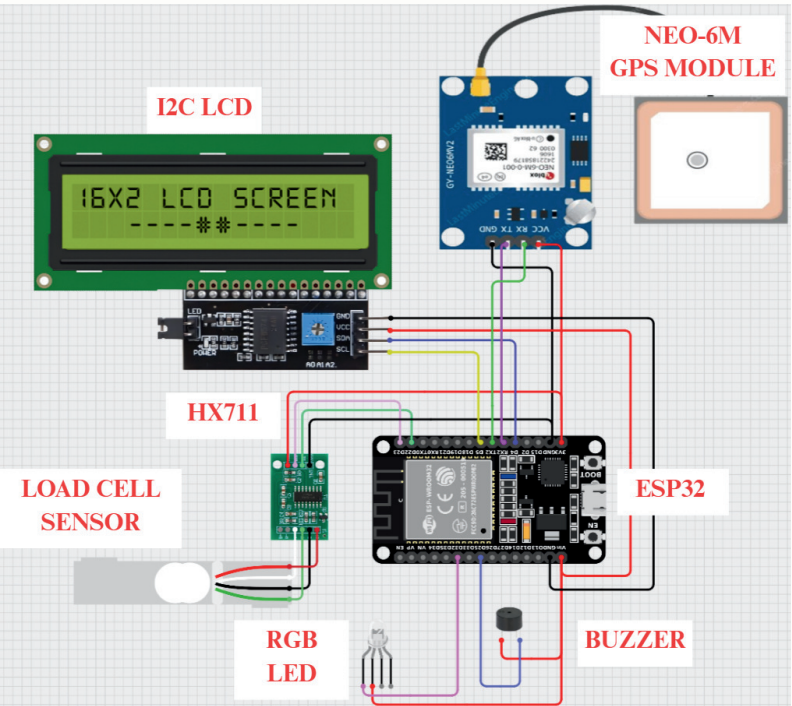
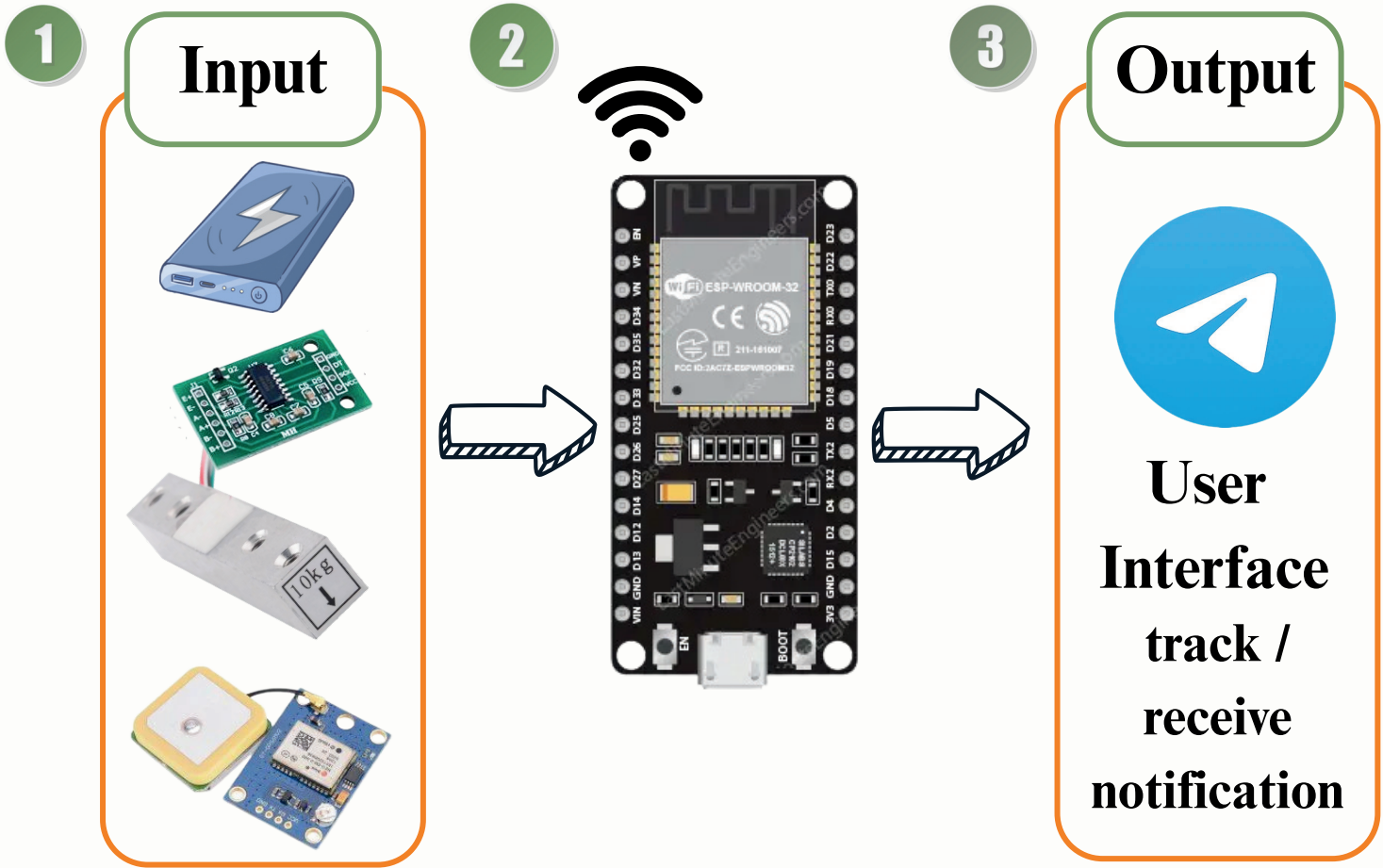


Introduction

The GPS-Based EduTrack Pack is a smart monitoring system that **MEASURE SCHOOL BAG WEIGHT** and **TRACKS STUDENT LOCATION IN REAL-TIME**. Using an ESP32, HX711, Load Cell, and GPS module, data is sent to parents via Telegram when the bag is too heavy or the child moves outside a safe zone. Developed in response to parental concerns, the project meets all objectives and offers a practical solution for improving student health and safety.



Methodology / Product Development



Conclusion

This project demonstrates an effective solution for monitoring school bag weight and student location to promote health, safety, and parental awareness.

Future improvements could include GPS module with higher connectivity and built-in Wi-Fi module to improve reliability and ease of use.



Objective



TRACK REAL-TIME LOCATION of students using GPS technology to improve parental monitoring.



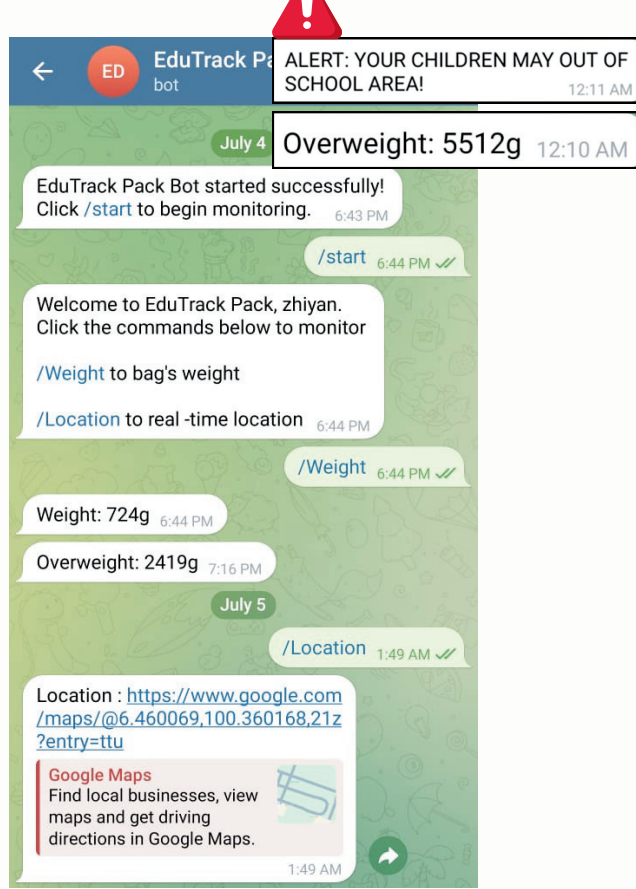
Measure the **REAL-TIME WEIGHT OF SCHOOL BAGS** using the HX711 with Load Cell Sensor to prevent physical strain on students.



PROVIDE INSTANT NOTIFICATIONS to parents via Telegram when the school bag weight exceeds safe limits or the student leaves a designated safe zone.



Result & Discussion



Environmental & Sustainability

SDG 3: GOOD HEALTH AND WELL-BEING

Promote student health by reducing risks of back pain and physical strain from heavy school bags.

SDG 4: QUALITY EDUCATION

Promotes safer learning by managing bag weight and enabling real-time parental monitoring.

Public Health, Safety, Culture



- Prevent **PHYSICAL STRAIN** in children
- Tracking real-time location for **SAFETY**
- **ALERT PARENTS** via Telegram

